



Code Luck

Build Your Lucky Future

Feature List & MVP Scope

Product Capabilities, RICE Prioritization & MVP Execution Strategy

1. Executive Summary & MVP Product Strategy

.Scora (developed by Team Code Luck under the slogan 'Build Your Lucky Future') is an AI-powered technical verification and talent discovery platform built specifically for software developers and enterprise hiring organizations. Scora solves the fundamental crisis in modern software engineering: the mass delivery of unverified, AI-generated code by developers who lack underlying architectural comprehension, memory management knowledge, or security awareness.

Modern product management research emphasizes extreme discipline during initial MVP scoping. Studies by Pendo (2019/2024 Feature Adoption Report) reveal that up to 80% of features in traditional B2B SaaS applications are rarely or never used, resulting in wasted engineering capital, bloated codebases, and product-market misalignment. Furthermore, research from the Standish Group (CHAOS Research) confirms that scope bloat is the primary driver of early-stage software failure.

Core MVP Hypothesis: Can Scora reliably evaluate a developer's technical engineering comprehension through dynamic AI challenges and automated AST maintainability analysis, creating a trusted, cryptographic Developer Passport that hiring teams actively use for candidate discovery?

2. Product Philosophy & Verification Principles

Scora's product architecture is built upon three non-negotiable core principles:

- 1. AI Tool Agnosticism: Scora is explicitly NOT anti-AI. We recognize LLM coding assistants (GitHub Copilot, Cursor, ChatGPT) as standard developer productivity tools. However, we enforce strict accountability by evaluating whether the developer truly understands, can explain, and can maintain the code delivered.
- 2. Comprehension Over Output Speed: Existing platforms measure raw code generation speed or LeetCode syntax memorization. Scora evaluates Abstract Syntax Tree (AST) complexity, cyclomatic maintainability ($C < 10$), DRY duplication index, and OWASP security clearance.
- 3. Behavioral Signal Analysis (Not Absolute AI Detection): Technical literature (METR 2024, Purdue 2023) proves that static AI code detectors suffer from high false-positive rates. Scora treats AI detection as behavioral telemetry signals (copy-paste entropy, keystroke dynamic cadence, focus loss) rather than absolute proof.

3. User Persona Specifications & JTBD Analysis

To ensure precise product-market alignment, Scora maps features directly to three core user personas:

User Persona	Job-to-be-Done (JTBD) Statement	Core Pain Points	Key Validation Triggers
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Software Developer (Supply Side)	When applying for remote engineering roles, I want an objective, verifiable way to prove my true technical comprehension, so that I can bypass resume filters and command premium rates (\$65-\$180/hr).	• Bids buried under 100+ low-cost prompt spammers. • Unfair star ratings on legacy gig platforms. • Repetitive technical interviews for every client.	• Cryptographic Developer Passport. • Objective Trust Score (0-1000). • Priority matching with high-budget enterprise clients.
Individual Client / Founder (Demand Side)	When hiring contract developers for early-stage software products, I want to verify that delivered code is secure and maintainable, so that I don't waste \$20k+ fixing technical debt.	• Receiving toxic, unmaintainable AI-bloated code. • Paying senior rates for hidden ChatGPT copy-pasting. • Post-launch security breaches and broken logic.	• 100% Code Maintainability Guarantee. • Instant Developer Passport audit view. • Verified skill breakdown.
Enterprise Hiring Team (B2B Side)	When scaling engineering teams across remote markets, I want an automated, high-trust candidate verification pipeline, so that I can reduce time-to-hire by 70% with zero code quality drop.	• High cost of manual engineer interviewing (\$5k+/hire). • High candidate cheat rates on LeetCode screening. • Resume inflation and unverified claims.	• Automated AST code quality report. • Anti-cheat behavioral telemetry logs. • Pre-verified talent search directory.

4. Complete Feature Inventory (20 Core Modules)

The Scora platform architecture comprises 20 comprehensive feature modules detailing over 100 specific sub-capabilities:

- 1. Authentication & Account Management: OAuth2 (GitHub, Google), JWT session tokens, Role-Based Access Control (Developer / Client / Enterprise), Email Verification, Multi-Factor Auth (MFA), Password Reset, Profile Deletion (GDPR).
- 2. Developer Experience & Profile System: Personalized Developer Profile, Verified Skills Matrix, Programming Languages, Frameworks, Experience History, Open-Source Portfolio, GitHub Sync, Interactive Dashboard, Assessment History, Skill Progress Graphs.
- 3. Assessment Engine & Challenge Library: Dynamic Assessment Library, Skill-Based Challenges, Junior-to-Staff Difficulty Scaling, AI Challenge Generator, Instructions Engine, Timed Execution Rules, Multi-Language Support (Rust, Go, Python, TypeScript, C++), Attempt & Retake Policies.
- 4. Live Code Execution & Browser IDE: Monaco Browser IDE (VS Code core), Syntax Highlighting, Isolated Sandbox Code Execution (gVisor/Docker), Public Test Case Runner, Hidden Test Cases, Real-time Console Output, Error Traceback Parsing, Auto-save (15s interval), Multi-file Project Support.
- 5. Abstract Syntax Tree (AST) Code Engine: Static Code Parsing (tree-sitter), Cyclomatic Complexity Calculation (McCabe Metric $C < 10$), Halstead Volume, Maintainability Index ($MI > 65$), DRY Duplication Scanner ($< 5\%$), OWASP Top 10 Security Audit, Automated Code Review Feedback Report.

- 6. Behavioral Telemetry & Trust Engine: Assessment Integrity Telemetry, Focus Loss / Tab-Switching Counter, Copy/Paste Payload Entropy Analysis, Keystroke Dynamic Cadence Tracking, Suspicious Behavior Signal Detection, Trust Event Logging, Bayesian Trust Score Calculation (0-1000), Historical Score Audit Trail.
- 7. Gamified Skill Points (SP) System: Skill Points Accumulation, Category-Specific SP Breakdown, Skill Tier Levels (Bronze, Silver, Gold, Platinum, Diamond), Skill Progress Velocity, Verified Skill Badging System.
- 8. Cryptographic Developer Passport: Decentralized Verifiable Credential (W3C Standard), Ed25519 Cryptographic Signatures, Trust Score Badge, SP Display, Verified Competencies List, Public Passport URL Generator, QR Code Verification, Passport Verification API, Granular Privacy Toggles.
- 9. Live AI Technical Interviewer: WebRTC Real-time Voice Stream, FastText NLP Engine, Dynamic Conversational Prompt Trees, Code Architecture Probing ('Why did you select this data structure?'), Interview Evaluation Scoring, Feedback Summary, Interview History.
- 10. Human Verification Peer Network: Double-Blind Peer Review Portal, Senior Engineer Reviewer Panel, Consensus Scoring Rubric, Reviewer Feedback Engine, Verification Approval / Rejection Workflow, Candidate Appeal Escalation.
- 11. Client Discovery & Search Portal: Client Onboarding, Public Developer Directory Search, Skill Filters, Minimum Trust Score Filter, Experience Tier Filter, Developer Comparison Engine, Saved Candidate Shortlists, Direct Contact Trigger, Project Brief Generator.
- 12. Company / Enterprise Hiring Pipeline: Company Profile Management, Team Workspaces, Enterprise Candidate Search, Advanced Custom Filters, Candidate Pipeline Kanban Board, Custom Technical Assessment Builder, Candidate Analytics Dashboard, Team Permissions Management.
- 13. Freelance Marketplace (Phase 3): Project Milestone Posting, Developer Proposals, Bid Management, Real-time Chat Messaging, Smart Contract Escrow, Escrow Release Triggers, Client Review System, Dispute Resolution Arbitration.
- 14. Multi-Modal AI Engine: OpenAI GPT-4o / Claude 3.5 Sonnet Integration via Instructor JSON Schemas, Dynamic Synthetic Challenge Generator, AI Assessment Feedback Generator, AI Skill Gap Diagnostics, AI Code Reviewer Bot.
- 15. Real-Time Platform Analytics: Developer Skill Velocity Analytics, Assessment Completion & Pass Rate Metrics, Trust Score Distribution Curve, Client Contact Conversion Rates, Enterprise Hiring Funnel Performance.
- 16. Notification & Alert System: In-App Notification Drawer, Email Dispatch System (SendGrid), Assessment Completion Alerts, Trust Score Update Alerts, Client Direct Outreach Notifications.
- 17. Administration & Governance Portal: Master Admin Dashboard, User Account Governance, Assessment Library Management, Challenge Generator Audit, Reviewer Payout Management, Abuse & Fraud Flagging Console, Platform Analytics.
- 18. API & Third-Party Integrations: GitHub Actions Integration (`scora-comprehension-check`), LinkedIn Credential Sync, Developer Passport Verification API, ATS Webhooks (Greenhouse, Lever), Enterprise Single Sign-On (Okta / SAML).
- 19. Security, Isolation & Compliance: Role-Based Access Control (RBAC), Isolated Container Execution Sandboxing, Network Ingress/Egress Blockers, Rate Limiting (Redis), Comprehensive Audit Logging, System Status Telemetry.

- 20. Data Privacy & Audit Logs: GDPR/CCPA Compliance Engine, Automated Data Deletion Pipeline, Immutable Audit Logs, Candidate Privacy Controls, Escrow Dispute Audit Trail.

5. Feature Prioritization Framework (RICE Analysis)

To rigorously justify MVP inclusion, every feature was evaluated using Intercom's RICE Prioritization Model:

RICE Score = (Reach × Impact × Confidence) / Effort

Where Reach = Users/Mo, Impact = 0.5 (Low) to 3.0 (Massive), Confidence = 50% to 100%, Effort = Person-Weeks.

Feature Category	Reach	Impact	Confidence	Effort	RICE Score	Prioritization Tier
Authentication & User Profile	100%	2.0 (High)	100%	2 wks	100.0	P0 — MVP Must Have
AI Dynamic Challenge Generator	100%	3.0 (Massive)	90%	3 wks	90.0	P0 — MVP Must Have
Monaco Browser IDE & Sandbox	100%	3.0 (Massive)	95%	4 wks	71.3	P0 — MVP Must Have
Automated AST Evaluation Engine	100%	3.0 (Massive)	90%	3 wks	90.0	P0 — MVP Must Have
Trust Engine v1 (Behavior Telemetry)	100%	2.5 (High)	85%	2 wks	106.3	P0 — MVP Must Have
Cryptographic Developer Passport	100%	3.0 (Massive)	90%	2 wks	135.0	P0 — MVP Must Have
Developer Search & Contact	80%	2.5 (High)	90%	2 wks	90.0	P0 — MVP Must Have
GitHub Action Badge (`scora-check`)	60%	2.0 (High)	80%	3 wks	32.0	P1 — Growth (Phase 2)
Live Voice AI Technical Interviewer	40%	2.5 (High)	70%	6 wks	11.7	P2 — Expansion (Phase 2)
Freelance Payment & Escrow Market	20%	3.0 (Massive)	60%	12 wks	3.0	P3 — Deferred (Phase 3)

6. Granular MVP Scope Definition

The Scora Phase 1 MVP is strictly bounded to validate technical evaluation and candidate discovery:

6.1 Developer MVP Boundary

- Included: Signup/Login, Profile Creation, Dynamic AI Assessment Execution, Live Monaco Code Editor, Sandbox Code Execution, Test Case Feedback, AST Quality Analysis, Trust Telemetry Tracking, Trust Score & SP Calculation, Developer Passport Display.
- Excluded: Live voice AI interviewing, GitHub repository auto-sync, peer review appeals.

6.2 Client / Company MVP Boundary

- Included: Client Registration, Public Developer Directory Search, Skill & Trust Score Filters, Developer Passport Detailed View, Direct Contact Form.
- Excluded: Project milestone contracts, Escrow payment handling, ATS webhooks, custom enterprise assessment builders.

7. System Architecture & Dependency Topology

Features interact through a strictly directed acyclic graph (DAG) execution sequence:

Sequential Architecture Dependency Flow:

1. Auth System → 2. Developer Profile → 3. Assessment Library → 4. AI Challenge Engine → 5. Monaco Browser IDE → 6. Containerized Code Sandbox → 7. AST Code Quality Engine → 8. Trust Telemetry Engine → 9. Trust Score & SP Calculator → 10. Developer Passport Generator → 11. Talent Search Portal → 12. Direct Client Outreach

Fail-Safe Fallback: If AST Engine encounters unparseable syntax, system falls back to automated test case execution scoring while flagging syntax anomalies for reviewer audit.

8. Exhaustive Feature Matrix Table

Feature Name	User Role	Priority	In MVP ?	Depends On	Connected To	Business Value	Risk Level
User Registration / OAuth	All Roles	P0	☑	None	User Profile	Access Security	Low
Developer Profile Setup	Developer	P0	☑	Auth System	Dashboard	User Onboarding	Low
AI Challenge Generator	Developer	P0	☑	Profile Setup	Monaco IDE	Assessment Engine	Moderate
Monaco Browser Code IDE	Developer	P0	☑	AI Challenge	Code Sandbox	Interactive Testing	Low
Isolated Sandbox Execution	System	P0	☑	Monaco IDE	AST Engine	Secure Code Audit	Moderate
AST Quality Analysis Engine	System	P0	☑	Code Sandbox	Trust Engine	Maintainability Audit	Moderate
Trust Telemetry Tracking	System	P0	☑	Monaco IDE	Trust Score	Anti-Cheat Guard	High
Trust Score Calculation	System	P0	☑	Trust Telemetry	Passport	Objective Credibility	Low
Skill Points (SP) Engine	Developer	P0	☑	AST Engine	Passport	User Gamification	Low
Developer Passport Display	Dev / Client	P0	☑	Trust Score	Talent Search	Verified Credential	Low
Developer Search Directory	Client / Co	P0	☑	Passport	Contact Form	Talent Discovery	Low
Trust Score & Skill Filters	Client / Co	P0	☑	Search Dir	Passport View	Precision Search	Low
Direct Client Contact Form	Client / Co	P0	☑	Search Dir	Email Alert	Lead Generation	Low
GitHub Action SP Badge	Developer	P1	—	Passport	GitHub Repo	Organic Growth	Moderate
Public Passport URL Sharing	Developer	P1	—	Passport	Social Media	Viral Acquisition	Low
Company Talent Workspace	Company	P1	—	Auth System	Search Dir	B2B SaaS Utility	Low
Live Voice AI Interviewer	Developer	P2	—	AST Engine	Trust Engine	Deep Probe	High
Double-Blind Peer Review	Dev / Reviewer	P2	—	AST Engine	Passport	Tier-3 Audit	Moderate

Freelance Payment Escrow	Dev / Client	P3	—	Search Dir	Contracts	Market Monetization	Critical
Smart Contract Milestones	Dev / Client	P3	—	Escrow Market	Disputes	Transaction Trust	Critical

9. User Stories & Acceptance Criteria Matrix

Comprehensive boundary conditions and acceptance criteria for major MVP modules:

9.1 AI Coding Challenge & Monaco IDE

• User Story: As a developer, I want to initiate a dynamic AI coding challenge in my browser editor, so that I can solve a real refactoring problem and prove my engineering skills.

1. System generates a dynamic problem statement, initial code repository, constraints, and visible unit tests within < 3 seconds.
2. Monaco IDE loads with language-specific syntax highlighting, auto-completion, and error linting.
3. User can execute visible unit tests inside an isolated sandbox container and receive stdout/stderr execution logs.
4. Code state is auto-saved locally every 15 seconds; timer auto-submits upon expiry.

9.2 AST Code Evaluation & Trust Engine Telemetry

• User Story: As the Scora platform, I want to analyze candidate code structure and behavioral signals during assessment, so that I can compute an objective Trust Score and maintain platform integrity.

1. AST Engine computes Cyclomatic Complexity (C < 10 target), DRY duplication percentage, and OWASP vulnerability clearance.
2. Trust Telemetry tracks focus loss (tab switches), copy/paste payload entropy, and keystroke dynamic cadence without invading user privacy.
3. System computes a composite Trust Score (0-1000) and updates the Developer Passport within < 5 seconds of submission.
4. If suspicious activity exceeds threshold (> 5 tab switches or high paste entropy), test is flagged for secondary review.

9.3 Talent Search & Developer Passport

• User Story: As a hiring manager, I want to filter verified developers by Trust Score and view their cryptographic Developer Passport, so that I can hire high-trust engineers with confidence.

1. Client search portal renders verified developers filtered by skill tags and minimum Trust Score thresholds.
2. Clicking candidate profile opens Developer Passport detailing verified skills, SP score, and AST code quality metrics.

3. Client can submit direct outreach message triggering verified email notification to candidate.

10. Business Model Alignment & Monetization Mapping

Features are strategically mapped to value creation across B2C (Developer) and B2B (Enterprise) segments:

Feature Module	Target User	Business Purpose & Value Creation	Monetization Point	Decision Status
Developer Passport	Developer + Company	Establishes technical credibility; drives organic developer onboarding.	Viral Supply Growth	Confirmed Product Decision
Trust Score Engine	Developer + Company	Eliminates hiring uncertainty; core moat against Upwork prompt debt.	Platform Differentiator	Confirmed Product Decision
Premium Verification Cert	Individual Developer	Unlocks verified Developer Passport issuance and priority matching.	\$49 one-time certification	Confirmed Product Decision
Company Hiring SaaS Sub	Enterprise Hiring Team	B2B SaaS dashboard access to search, filter, and contact verified talent.	\$299 - \$999 / month sub	Confirmed Product Decision
Developer Vetting API	Corporate HR / ATS	API calls executing automated AI comprehension tests on applicants.	\$0.50 per assessment call	Proposed Assumption
Escrow Freelance Market	Dev + Client	Milestone commission take-rate on completed contract payments.	15% marketplace take-rate	Phase 3 Proposed Assumption

11. Out-of-Scope Rationale & Risk Analysis

The following major capabilities are explicitly excluded from Phase 1 MVP to mitigate launch risk:

- **1. Freelance Payment & Escrow System:** Implementing escrow payments, milestone holding, and chargeback arbitration requires complex money-transmitter licensing and Stripe Connect integration. Postponing escrow to Phase 3 allows Scora to validate developer verification without regulatory friction.
- **2. Live Voice AI Technical Interviewer:** WebRTC voice streaming and real-time LLM inference incur higher latency and token costs. Postponing voice interviewing to Phase 2 focuses initial R&D; on AST code maintainability models.
- **3. Enterprise SSO & ATS Integrations:** Okta/SAML integrations and Greenhouse/Lever webhooks require long enterprise sales cycles. Postponing ATS integrations to Phase 2 ensures MVP focus on core talent search.

12. Quantitative MVP Success Metrics

MVP performance will be evaluated against objective quantitative KPI targets:

Metric Category	Specific KPI Indicator	Measurement Formula / Method	MVP Target Benchmark
Developer Engagement	Assessment Completion Rate	$(\text{Completed Assessments} / \text{Started Assessments}) \times 100$	> 75% Completion Rate
Developer Activation	Passport Creation Rate	$(\text{Verified Passports Created} / \text{Total Registrations}) \times 100$	> 60% Activation Rate
Trust Integrity	False Positive Flag Rate	$(\text{Incorrect Cheating Flags} / \text{Total Telemetry Audits}) \times 100$	< 3.0% False Positive Rate
Client Conversion	Search-to-Contact Rate	$(\text{Developer Outreach Messages Sent} / \text{Passport Views}) \times 100$	> 12.5% Contact Rate
System Performance	Code Evaluation Latency	Time from code submission to AST report generation	< 4.5 Seconds Latency

13. 36-Month Feature Expansion Roadmap

- Phase 1 MVP (Months 0 - 6): Core AI Challenge Engine, Monaco IDE, Sandbox Execution, AST Quality Engine, Trust Engine v1, Developer Passport, Talent Search Portal.
- Phase 2 Network Scale (Months 6 - 12): Live Voice AI Technical Interviewer, GitHub Action SP Badges (`scora-check`), Double-Blind Peer Review Network, B2B Vetting API.
- Phase 3 Enterprise & Marketplace (Months 12 - 24): Custom Fine-Tuned AST Code Reviewer AI, Freelance Payment Escrow, Smart Contract Milestones, Enterprise ATS Integrations.
- Phase 4 Global Dominance (Months 24 - 36): Cross-platform Developer Passport global standard, localized multi-lingual AI interviewers, automated escrow code audits.

14. Technical & Academic Bibliography

1. Pendo (2019/2024): 'Feature Adoption Report: 80% of SaaS Features Rarely or Never Used'. Source: pendo.io/data-reports
2. Standish Group (2024): 'CHAOS Study on Software Project Scope & Feature Bloat'. Source: standishgroup.com
3. GitClear (2024-2025): 'Coding on Copilot: Longitudinal Analysis of 153M Lines of Code' (7.1% churn, 8x duplicate code). Source: gitclear.com
4. Stanford University (Perry et al., ACM CCS 2023): 'Do Users Write More Insecure Code with AI Assistants?' (Overconfidence Paradox study). Source: arxiv.org/abs/2211.03622
5. CISQ National Report (2024): 'Cost of Poor Software Quality in the U.S. Report' (\$2.41T impact). Source: it-cisq.org
6. Purdue University (2023): 'Analysis of ChatGPT SE Answers' (52% error rate). Source: arxiv.org/abs/2308.02312
7. JetBrains (2025): 'State of Developer Ecosystem Survey' (24,534 developers). Source: jetbrains.com
8. SVPG / Marty Cagan (2024): 'Inspired: How to Create Tech Products Customers Love (Product Discovery Framework)'. Source: svpg.com

9. W3C Verifiable Credentials Data Model v2.0 (2024): Developer Passport standard. Source: w3.org/TR/vc-data-model-2.0
10. IEEE Transactions on Software Engineering (2024): AST cyclomatic complexity (McCabe C < 10). Source: [ieee.org](https://ieeexplore.ieee.org)
11. OWASP Foundation (2024-2025): 'OWASP Top 10 for LLM Applications & Security Standards'. Source: owasp.org
12. Upwork & Fiverr SEC Form 10-K (2025): Financial scale and platform volume benchmarks. Source: [sec.gov](https://www.sec.gov)